

STANDARD 2

Stage 6

Financial Maths (Std 2), F4 Investments and Loans (Y12)

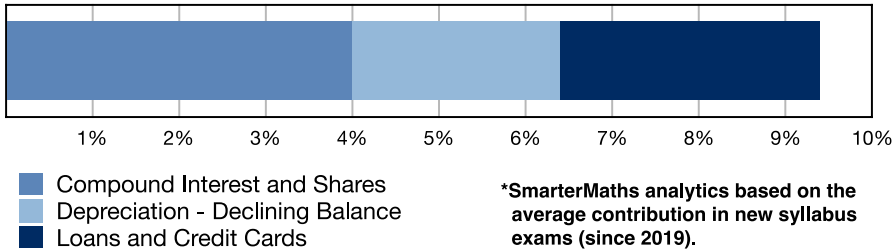
Compound Interest and Shares (Std2)

Teacher: Hailey Summer

Exam Equivalent Time: 84 minutes (based on allocation of 1.5 minutes per mark)

STD2 Exam Contribution History

F4 - Investments and Loans



IMPORTANT FEATURES AND TIPS FROM EXAM HISTORY

- MS-F4 Investments and Loans* has contributed a substantial 9.4% to the new syllabus Std2 exams since its introduction in 2019.
- We have split this area into 3 sub-topics for analysis purposes: *Compound Interest and Shares* (4.0%), *Depreciation - Declining Balance* (2.4%) and *Loans and Credit Cards* (3.0%).
- This analysis looks at the largest sub-topic *Compound Interest and Shares* (4.0%).

ANALYSIS - What to Expect and Common pitfalls

- Compound Interest* consistently causes problems, particularly with questions involving *Compounded Value of \$1* tables which have been asked 5 times since 2011 (surprisingly absent in new syllabus exams to date), producing sub-50% mean marks on 4 occasions.
- Examiners have required students to use the **$FV = PV(1 + r)^n$** formula 8 times in the last 10 years (most recently in 2023). A regularly tested topic that is overdue for a more substantial mark allocation.
- 2021 Std2 Q26b* required a novel comparison between compound and simple interest that caused problems for a majority of students and should be carefully reviewed.
- Dividend Yield* was not examined in the 2023 HSC exam, breaking a trend of four years in a row. We expect this challenging topic area to be consistently examined going forward and highly recommend a revision focus.

Questions

1. Financial Maths, STD2 F4 2009 HSC 6 MC

A house was purchased in 1984 for \$35 000. Assume that the value of the house has increased by 3% per annum since then.

Which expression gives the value of the house in 2009?

- A. $35\,000(1 + 0.03)^{25}$
- B. $35\,000(1 + 3)^{25}$
- C. $35\,000 \times 25 \times 0.03$
- D. $35\,000 \times 25 \times 3$

2. Financial Maths, STD2 F4 2022 HSC 10 MC

Alex purchased 800 shares. The total cost was \$2.60 per share. Alex sold the shares one year later for \$3.40 each and paid a fee of \$24.95 for selling the shares.

What profit did Alex make on these shares?

- A. \$590.10
- B. \$615.05
- C. \$640.00
- D. \$664.95

3. Financial Maths, STD2 F4 2015 HSC 17 MC

What amount must be invested now at 4% per annum, compounded quarterly, so that in five years it will have grown to \$60 000?

- A. \$8919
- B. \$11 156
- C. \$49 173
- D. \$49 316

4. Financial Maths, STD2 F4 2016 HSC 8 MC

The table shows the future value of an investment of \$1000, compounding yearly, at varying interest rates for different periods of time.

Future values of an investment of \$1000					
Number of years	Interest rate per annum				
	1%	2%	3%	4%	5%
1	1010.00	1020.00	1030.00	1040.00	1050.00
2	1020.10	1040.40	1060.90	1081.60	1102.50
3	1030.30	1061.21	1092.73	1124.86	1157.63
4	1040.60	1082.43	1125.51	1169.86	1215.51
5	1051.01	1104.08	1159.27	1216.65	1276.28

Based on the information provided, what is the future value of an investment of \$2500 over 3 years at 4% pa?

- A. \$1124.86
- B. \$2812.15
- C. \$3624.86
- D. \$5312.15

5. Financial Maths, STD2 F4 2017 HSC 10 MC

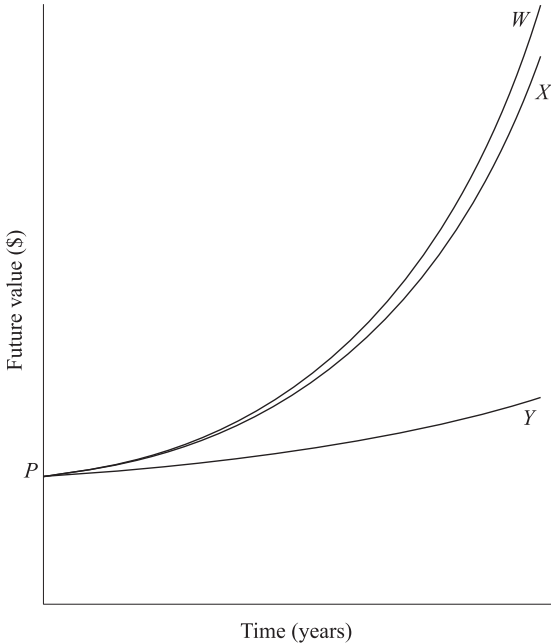
A single amount of \$10 000 is invested for 4 years, earning interest at the rate of 3% per annum, compounded monthly.

Which expression will give the future value of the investment?

- A. $10\,000 \times (1 + 0.03)^4$
- B. $10\,000 \times (1 + 0.03)^{48}$
- C. $10\,000 \times \left(1 + \frac{0.03}{12}\right)^4$
- D. $10\,000 \times \left(1 + \frac{0.03}{12}\right)^{48}$

6. Financial Maths, STD2 F4 2019 HSC 13 MC

The graph show the future values over time of **\$P**, invested at three different rates of compound interest.



Which of the following correctly identifies each graph?

- A.

W	5% pa, compounding annually
X	10% pa, compounding annually
Y	10% pa, compounding quarterly
- B.

W	5% pa, compounding annually
X	10% pa, compounding quarterly
Y	10% pa, compounding annually
- C.

W	10% pa, compounding quarterly
X	10% pa, compounding annually
Y	5% pa, compounding annually
- D.

W	10% pa, compounding annually
X	10% pa, compounding quarterly
Y	5% pa, compounding annually

7. Financial Maths, STD2 F4 2019 HSC 3 MC

Chris opens a bank account and deposits \$1000 into it. Interest is paid at 3.5% per annum, compounding annually.

Assuming no further deposits or withdrawals are made, what will be the balance in the account at the end of two years?

- A. \$1070.00
- B. \$1071.23
- C. \$1822.50
- D. \$2070.00

8. Financial Maths, STD2 F4 2020 HSC 4 MC

Joan invests \$200. She earns interest at 3% per annum, compounded monthly.

What is the future value of Joan's investment after 1.5 years?

- A. \$209.07
- B. \$209.19
- C. \$279.51
- D. \$311.93

9. Financial Maths, STD2 F2 2021 HSC 5 MC

Peter currently earns \$21.50 per hour. His hourly wage will increase by 2.1% compounded each year for the next four years.

What will his hourly wage be after four years?

- A. $21.50(1.21)^4$
- B. $21.50(1.021)^4$
- C. $21.50 + 21.50 \times 0.21 \times 4$
- D. $21.50 + 21.50 \times 0.021 \times 4$

10. Financial Maths, STD2 F4 2022 HSC 11 MC

In ten years, the future value of an investment will be \$150 000. The interest rate is 4% per annum, compounded half-yearly.

Which equation will give the present value (***PV***) of the investment?

- A. $PV = \frac{150\,000}{(1 + 0.04)^{10}}$
- B. $PV = \frac{150\,000}{(1 + 0.04)^{20}}$
- C. $PV = \frac{150\,000}{(1 + 0.02)^{10}}$
- D. $PV = \frac{150\,000}{(1 + 0.02)^{20}}$

11. Financial Maths, STD2 F4 2023 HSC 10 MC

An amount of \$25 000 is invested for six years. Interest is earned at a rate of 8% per annum, compounding quarterly.

Which expression gives the value of the investment after 6 years, in dollars?

- A. $25\,000 \times 1.02^{24}$
- B. $25\,000 \times 1.02^6$
- C. $25\,000 \times 1.08^{24}$
- D. $25\,000 \times 1.08^6$

12. Financial Maths, STD2 F4 2024 HSC 7 MC

Three years ago, the price of a uniform was \$180.

Due to inflation, the price increased annually by 2.5%.

What is the price of this uniform now?

- A. \$180.14
- B. \$ 181.35
- C. \$ 193.50
- D. \$ 193.84

13. Financial Maths, STD2 F4 2012 HSC 9 MC

Tracy invests some money for 2 years at 4% per annum, compounded quarterly.

Compounded values of \$1					
Period	Interest rate per period				
	1%	2%	3%	4%	5%
1	1.010	1.020	1.030	1.040	1.050
2	1.020	1.040	1.061	1.082	1.103
3	1.030	1.061	1.093	1.125	1.158
4	1.041	1.082	1.126	1.170	1.216
5	1.051	1.104	1.159	1.217	1.276
6	1.062	1.126	1.194	1.265	1.340
7	1.072	1.149	1.230	1.316	1.407
8	1.083	1.172	1.267	1.369	1.477

Which figure from the table should Tracy use to calculate the value of her investment at the end of 2 years?

- A. 1.020
- B. 1.082
- C. 1.083
- D. 1.369

14. Financial Maths, STD2 F4 2005 HSC 13 MC

Last year, Helen bought 150 shares at \$2.00 per share. They are now worth \$2.50 per share. Helen receives a dividend of \$0.10 per share.

What is the dividend yield?

- A. 4%
- B. 20%
- C. \$15
- D. \$75

15. Financial Maths, STD2 F4 2018 HSC 19 MC

The table shows the compounded values of \$1 at different interest rates over different periods.

Compounded values of \$1					
Number of periods	Interest rate per period				
	1%	2%	3%	4%	5%
2	1.0201	1.0404	1.0609	1.0816	1.1025
4	1.0406	1.0824	1.1255	1.1699	1.2155
6	1.0615	1.1262	1.1941	1.2653	1.3401
8	1.0829	1.1717	1.2668	1.3686	1.4775
10	1.1046	1.2190	1.3439	1.4802	1.6289
12	1.1268	1.2682	1.4258	1.6010	1.7959

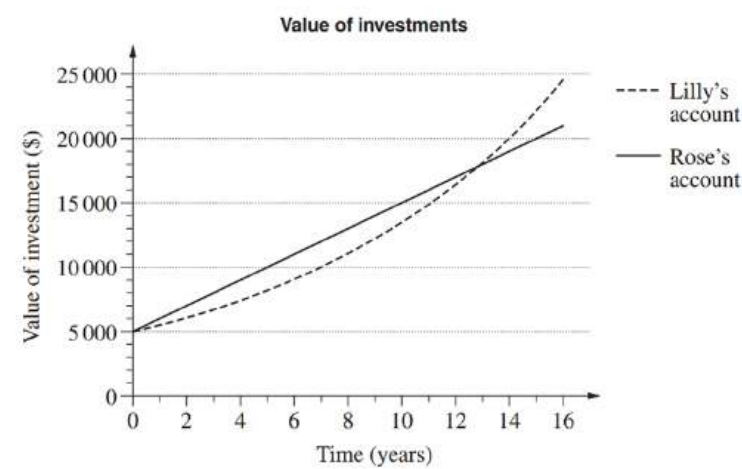
Amy hopes to have \$21 000 in 2 years to buy a car. She opens an account today which pays interest of 4% pa, compounded quarterly.

Using the table, which expression calculates the minimum single sum that Amy needs to invest today to ensure she reaches her savings goal?

- A. $21\,000 \times 1.0816$
- B. $21\,000 \div 1.0816$
- C. $21\,000 \times 1.0829$
- D. $21\,000 \div 1.0829$

16. Financial Maths, STD2 F1 2007 HSC 23a

Lilly and Rose each have money to invest and choose different investment accounts.
The graph shows the values of their investments over time.



- i. How much was Rose’s original investment? (1 mark)
- ii. At the end of 6 years, which investment will be worth the most and by how much? (2 marks)
- iii. Lilly’s investment will reach a value of \$20 000 first.
How much longer will it take Rose’s investment to reach a value of \$20 000? (1 mark)

17. Financial Maths, STD2 F4 2008 HSC 24c

Heidi’s funds in a superannuation scheme have a future value of \$740 000 in 20 years time. The interest rate is 4% per annum and earnings are calculated six-monthly.
What single amount could be invested now to produce the same result over the same period of time at the same interest rate? (3 marks)

18. Financial Maths, STD2 F4 2015 HSC 26d

A family currently pays \$320 for some groceries.
Assuming a constant annual inflation rate of 2.9%, calculate how much would be paid for the same groceries in 5 years’ time. (2 marks)

19. Financial Maths, STD2 F4 2017 HSC 26e

Sam purchased 500 company shares at \$3.20 per share. Brokerage fees were 1.5% of the purchase price.
Sam is paid a dividend of 26 cents per share, then immediately sells the shares for \$4.80 each.
If he pays no further brokerage fees, what is Sam’s total profit? (3 marks)

20. Financial Maths, STD1 F2 2021 HSC 14

It costs \$2.45 for a car to travel on a toll road. Due to inflation, the cost is to increase by 3% each year.
How much will it cost for a car to travel on the toll road in 5 years time? (2 marks)

21. Financial Maths, STD2 F4 2020 HSC 21

The inflation rate over the year from January 2019 to January 2020 was 2%.
The cost of a school jumper in January 2020 was \$122.
Calculate the cost of the jumper in January 2019 assuming that the only change in the cost of the jumper was due to inflation. (2 marks)

22. Financial Maths, STD2 F4 2019 HSC 21

A person owns 1526 shares with a market value of \$8.75 per share. The total dividend received for these shares is \$1068.20.
Calculate the percentage dividend yield. (2 marks)

23. Financial Maths, STD2 F4 2024 HSC 25

Alex and Jun each invest \$1800 for 5 years.

- Alex's investment earns simple interest at a rate of 7.5% per annum.
- Jun's investment earns interest at a rate of 6.0% per annum, compounding quarterly.

By calculating the interest earned over the 5 years, determine who will have the greater amount. (3 marks)

24. Financial Maths, STD2 F4 2021 HSC 26

Nina plans to invest \$35 000 for 1 year. She is offered two different investment options.

Option A: Interest is paid at 6% per annum compounded monthly.

Option B: Interest is paid at r % per annum simple interest.

- a. Calculate the future value of Nina's investment after 1 year if she chooses Option A. (2 marks)
- b. Find the value of r in Option B that would give Nina the same future value after 1 year as for Option A.
Give your answer correct to two decimal places. (2 marks)

25. Financial Maths, STD2 F4 SM-Bank 1

An investment fund purchases 4500 shares of Bank ABC for a total cost of \$274 500 (ignore any transaction costs).
The investment fund is paid a divided of \$3.66 per share in the first year.

- i. What was the purchase price of 1 share? (1 mark)
- ii. Calculate the divided yield. (1 mark)

26. Financial Maths, STD2 F4 2014 HSC 30a

Chandra and Sascha plan to have \$20 000 in an investment account in 15 years time for their grandchild's university fees.

The interest rate for the investment account will be fixed at 3% per annum compounded monthly.

Calculate the amount that they will need to deposit into the account now in order to achieve their plan. (3 marks)

27. Financial Maths, STD2 F4 2011 HSC 23c

An amount of \$5000 is invested at 10% per annum, compounded six-monthly.

Compounded values of \$1					
Period	Interest rate per period				
	1%	5%	10%	15%	20%
1	1.010	1.050	1.100	1.150	1.200
2	1.020	1.103	1.210	1.323	1.440
3	1.030	1.158	1.331	1.521	1.728
4	1.041	1.216	1.464	1.750	2.074
5	1.051	1.276	1.611	2.011	2.488
6	1.062	1.340	1.772	2.313	2.986

Use the table to find the value of this investment at the end of three years. (2 marks)

28. Financial Maths, STD2 F4 2013 HSC 26e

Kimberley has invested \$3500.

Interest is compounded half-yearly at a rate of 2% per half-year.

Compounded values of \$1						
Period	Interest rate per period					
	1%	2%	3%	4%	5%	6%
1	1.010	1.02	1.03	1.04	1.05	1.06
2	1.020	1.040	1.061	1.082	1.103	1.124
3	1.030	1.061	1.093	1.125	1.158	1.191
4	1.041	1.082	1.126	1.170	1.216	1.262
5	1.051	1.104	1.159	1.217	1.276	1.338
6	1.062	1.126	1.194	1.265	1.340	1.419
7	1.072	1.149	1.230	1.316	1.407	1.504
8	1.083	1.172	1.267	1.369	1.477	1.594

Use the table to calculate the value of her investment at the end of 4 years. (2 marks)

29. Financial Maths, STD2 F4 2020 HSC 29

Jana owns a share portfolio. Details of her share portfolio at 30 June 2020 are given in the table.

Company name	Numbers of shares in Jana's portfolio	Dividend yield (per annum)	Market price per share
ABC	200	6.0 %	\$5.50
XYZ	?	4.0 %	\$6.00

Jana received a total annual dividend of \$149.52 from her share portfolio.

Calculate the number of shares Jana has in company **XYZ** on 30 June 2020. (3 marks)

30. Financial Maths, STD2 F4 2021 HSC 30

Ariana owns 1500 shares in a company. The market price for each share is \$27. Ariana's total dividend from these shares is \$810.

Calculate the dividend yield for her shares. (2 marks)

31. Financial Maths, STD2 F4 2013 HSC 28d

Adhele has 2000 shares. The current share price is \$1.50 per share. Adhele is paid a dividend of \$0.30 per share.

i. What is the current value of her shares? (1 mark)

ii. Calculate the dividend yield. (1 mark)

Worked Solutions

1. Financial Maths, STD2 F4 2009 HSC 6 MC

$$r = 3\% = 0.03$$

$$n = 25 \text{ years}$$

$$\text{Using } FV = PV(1 + r)^n$$

$$\therefore \text{Value in 2009} = 35\,000(1 + 0.03)^{25}$$

$$\Rightarrow A$$

2. Financial Maths, STD2 F4 2022 HSC 10 MC

$$\text{Cost of shares} = 800 \times 2.60 = \$2080$$

$$\text{Sale funds} = \text{sale price} - \text{fee}$$

$$= (800 \times 3.40) - 24.95$$

$$= 2695.05$$

$$\text{Profit} = \text{Sale funds} - \text{Cost}$$

$$= 2695.05 - 2080$$

$$= \$615.05$$

$$\Rightarrow B$$

3. Financial Maths, STD2 F4 2015 HSC 17 MC

$$\text{Using } FV = PV(1 + r)^n$$

$$r = \frac{4\%}{4} = 1\% = 0.01 \text{ per quarter}$$

$$n = 5 \times 4 = 20 \text{ quarters}$$

$$60\,000 = PV(1 + 0.01)^{20}$$

$$\therefore PV = \frac{60\,000}{1.01^{20}}$$

$$= \$49\,172.66\dots$$

$$\Rightarrow C$$

4. Financial Maths, STD2 F4 2016 HSC 8 MC

$$\text{Table factor} = 1124.86$$

$$\therefore FV = 2.5 \times 1124.86$$

$$= \$2812.15$$

$$\Rightarrow B$$

5. Financial Maths, STD2 F4 2017 HSC 10 MC

$$\text{Compounding rate} = \frac{3}{100} \div 12 = \frac{0.03}{12}$$

$$\text{Compounding periods} = 4 \times 12 = 48$$

$$\therefore FV = 10\,000 \times \left(1 + \frac{0.03}{12}\right)^{48}$$

$$\Rightarrow D$$

6. Financial Maths, STD2 F4 2019 HSC 13 MC

Values increase quicker

- higher compounding interest rate

- same rate but more frequent compounding period

$$\therefore W = 10\% \text{ quarterly}$$

$$X = 10\% \text{ annually}$$

$$Y = 5\% \text{ annually}$$

$$\Rightarrow C$$

7. Financial Maths, STD2 F4 2019 HSC 3 MC

$$FV = PV(1 + r)^n$$

$$= 1000(1 + 0.035)^2$$

$$= \$1071.23$$

$$\Rightarrow B$$

8. Financial Maths, STD2 F4 2020 HSC 4 MC

$$\text{Monthly interest rate} = \frac{0.03}{12}$$

$$n = 1.5 \times 12 = 18$$

$$FV = PV(1 + r)^n$$

$$= 200 \left(1 + \frac{0.03}{12}\right)^{18}$$

$$= \$209.19$$

$$\Rightarrow B$$

9. Financial Maths, STD2 F2 2021 HSC 5 MC

Wage after 1 year = 21.50×1.021
Wage after 2 years = $21.50 \times 1.021 \times 1.021 = 21.50(1.021)^2$
 $\vdots$
Wage after 4 years = $21.50(1.021)^4$
 $\Rightarrow B$

10. Financial Maths, STD2 F4 2022 HSC 11 MC

Compounding periods = $10 \times 2 = 20$
Compounding rate = $\frac{4\%}{2} = 2\% = 0.02$
 $PV = \frac{150\,000}{(1 + 0.02)^{20}}$
 $\Rightarrow D$

11. Financial Maths, STD2 F4 2023 HSC 10 MC

Interest rate = $\frac{8}{2} = 2\%$ per quarter
Compounding periods = $6 \times 4 = 24$
 $\therefore FV = 25\,000 \times 1.02^{24}$
 $\Rightarrow A$

12. Financial Maths, STD2 F4 2024 HSC 7 MC

$r = 2.5\% = \frac{2.5}{100} = 0.025$
 $FV = PV(1 + r)^n$
 $= 180(1.025)^3$
 $= 193.84$
 $\Rightarrow D$

13. Financial Maths, STD2 F4 2012 HSC 9 MC

4% annual
 $\Rightarrow \frac{4\%}{4} = 1\%$ compounded each quarter
 $\Rightarrow n = 8$ (8 quarters in 2 years)

 $\therefore$ Factor = 1.083 (from table)
 $\Rightarrow C$

◆◆◆ Mean mark 24%. The lowest MC mean mark in the 2012 exam.
COMMENT: A process to follow: 1- convert the annual rate to the rate per compounding period. 2-calculate the number of compounding periods.

14. Financial Maths, STD2 F4 2005 HSC 13 MC

Dividend yield = $\frac{\text{Dividend}}{\text{Share Value}}$
 $= \frac{0.10}{2.50}$
 $= 0.04$
 $= 4\%$
 $\Rightarrow A$

15. Financial Maths, STD2 F4 2018 HSC 19 MC

4% annual
 $\Rightarrow \frac{4\%}{4} = 1\%$ compounded quarterly
 $\Rightarrow n = 8$
 $\Rightarrow$ Factor = 1.0829
 $\therefore$ Minimum sum = $21\,000 \div 1.0829$
 $\Rightarrow D$

◆◆ Mean mark 33%.

16. Financial Maths, STD2 F1 2007 HSC 23a

- i. \$5000 (*y*-intercept)

ii. After 6 years,

Lilly's investment = \$9000
Rose's investment = \$11 000
 $\therefore$ Rose's is worth \$2000 more.

iii. It takes Lilly 14 years to reach \$20 000 and it takes Rose 1 year longer (15 years) to reach the same value.

17. Financial Maths, STD2 F4 2008 HSC 24c

$FV = PV(1 + r)^n$
 $740\,000 = PV\left(1 + \frac{2}{100}\right)^{40}$
 $\therefore PV = \frac{740\,000}{(1.02)^{40}}$
 $= 335\,138.907\dots$
 $= \$335\,138.91$

18. Financial Maths, STD2 F4 2015 HSC 26d

$$\begin{aligned}FV &= PV(1 + r)^n \\&= 320(1.029)^5 \\&= \$369.1703\dots \\&= \$369.17 \text{ (nearest cent)}\end{aligned}$$

19. Financial Maths, STD2 F4 2017 HSC 26e

$$\begin{aligned}\text{Purchase price} &= 500 \times 3.20 \\&= \$1600 \\ \text{Brokerage fees} &= 1.5\% \times 1600 \\&= \$24 \\ \text{Dividends} &= 500 \times 0.26 \\&= \$130 \\ \text{Sale price} &= 500 \times 4.80 \\&= \$2400 \\ \therefore \text{Total profit} &= 2400 + 130 - (1600 + 24) \\&= \$906\end{aligned}$$

20. Financial Maths, STD1 F2 2021 HSC 14

$$\begin{aligned}FV &= PV(1 + r)^n \\&= 2.45 \left(1 + \frac{3}{100}\right)^5 \\&= 2.45(1.03)^5 \\&= \$2.84\end{aligned}$$

21. Financial Maths, STD2 F4 2020 HSC 21

$$\begin{aligned}FV &= PV(1 + r)^n \\122 &= C_{2019}(1 + 0.02)^1 \\C_{2019} \times 1.02 &= 122 \\C_{2019} &= \frac{122}{1.02} \\&= \$119.61\end{aligned}$$

22. Financial Maths, STD2 F4 2019 HSC 21

$$\begin{aligned}\text{Value of shares} &= 1526 \times 8.75 \\&= \$13\,352.50 \\ \therefore \text{Dividend yield} &= \frac{1068.20}{13\,352.50} \\&= 0.08 \\&= 8\%\end{aligned}$$

23. Financial Maths, STD2 F4 2024 HSC 25

$$\begin{aligned}\text{Alex's investment:} \\ \text{Interest} &= Prn = 1800 \times 0.075 \times 5 = \$675 \\ \text{Jun's investment:} \\ r &= \frac{6.0\%}{4} = 1.5\% \text{ per quarter} \\ \text{Compounding periods} &= 5 \times 4 = 20 \\ FV &= PV(1 + r)^n = 1800(1 + 0.015)^{20} = \$2424.34 \\ \text{Total interest} &= FV - PV = 2424.34 - 1800 = \$624.34\end{aligned}$$

Alex's interest > Jun's interest.
⇒ Alex will have a greater amount (since original investment the same)

24. Financial Maths, STD2 F4 2021 HSC 26

$$\begin{aligned}\text{a. } r &= \frac{6\%}{12} = 0.5\% = 0.005 \text{ per month} \\ n &= 12\end{aligned}$$

$$\begin{aligned}FV &= PV(1 + r)^n \\&= 35\,000(1 + 0.005)^{12} \\&= \$37\,158.72\end{aligned}$$

$$\begin{aligned}\text{b. } I &= Prn \\ 2158.72 &= 35\,000 \times r \times 1 \\ r &= \frac{2158.72}{35\,000} \\&= 0.06167\dots \\&= 6.17\% \text{ (to 2 d.p.)}\end{aligned}$$

♦♦ Mean mark part (b) 36%.

25. Financial Maths, STD2 F4 SM-Bank 1

i. Price per share

$$\begin{aligned} &= \frac{274\,500}{4500} \\ &= \$61 \end{aligned}$$

ii. Dividend Yield

$$\begin{aligned} &= \frac{\text{Dividend}}{\text{Share Price}} \times 100 \\ &= \frac{3.66}{61} \times 100 \\ &= 6\% \end{aligned}$$

26. Financial Maths, STD2 F4 2014 HSC 30a

$$FV = \$20\,000, \quad n = 15 \times 12 = 180,$$

$$r = \frac{0.03}{12} = 0.0025$$

$$FV = PV(1 + r)^n$$

$$20\,000 = PV(1 + 0.0025)^{180}$$

$$\begin{aligned} PV &= \frac{20\,000}{(1.0025)^{180}} \\ &= 12\,759.73\dots \end{aligned}$$

∴ They need to deposit \$12 760 (nearest \$)

27. Financial Maths, STD2 F4 2011 HSC 23c

Interest rate = 10% pa = 5% per 6 months

Period = 6 (6 x 6 months in 3 years)

⇒ Table value = 1.340

∴ Value of investment = 5000×1.34
= \$6700

♦ Mean mark 49%

♦♦ Mean mark 28%
MARKER'S COMMENT: Remember that the number of periods is the number of "compounding periods" and when asked to use the table, use the table!

28. Financial Maths, STD2 F4 2013 HSC 26e

$r = 2\%$ per half-year

$n = 8$ (8 half-years in 4 years)

⇒ Table Factor = 1.172

Investment = 3500×1.172
= \$4102

∴ After 4 years, investment value is \$4102

♦ Mean mark 44%
COMMENT: Structure your answer:
1-Find the interest rate per compounding period (same in this case). 2-Find the number of compounding periods.

29. Financial Maths, STD2 F4 2020 HSC 29

ABC Dividend = Value of shares $\times$ dividend yield
= $(200 \times 5.5) \times 0.06$
= \$66.00

XYZ Dividend = $149.52 - 66.00$
= \$83.52

♦ Mean mark 43%.

Let x = number of XYZ shares

$83.52 = \text{Value of } XYZ \text{ shares} \times \text{dividend yield}$

$83.52 = (x \times 6.0) \times 0.04$

$$6x = \frac{83.52}{0.04}$$

$$\therefore x = \frac{2088}{6}$$

$$= 348$$

30. Financial Maths, STD2 F4 2021 HSC 30

Dividend yield = $\frac{\text{Dividend}}{\text{Value of shares}}$
= $\frac{810}{1500 \times 27}$
= 0.02
= 2% yield

♦ Mean mark 47%.

31. Financial Maths, STD2 F4 2013 HSC 28d

i. $\# \text{ Shares} = 2000$

$\text{Share price} = \$1.50$

$\therefore \text{Current value} = 2000 \times 1.50$
 $= \$3000$

ii. $\text{Dividend yield} = \frac{\text{Dividend}}{\text{Share price}}$
 $= \frac{0.30}{1.50}$
 $= 20\%$

◆◆◆ Mean mark 13%
MARKER'S COMMENT: A large majority of students had a poor understanding of the term dividend yield.